

QPLANT Cryogenic Dashboard — Recursive Build Log

Version: 4.0.0
Generated: 2026-05-14T12:00:00Z
Format: Chat-based recursive sequence with SHA-256 provenance

Build Genealogy

User Request (comprehensive implementation)

↓

[Phase 0] Setup & Audit → config.yaml [f49b9565cee0]

↓

[Phase 1] Immediate Fixes → index.html [93b94be27890], NAVIGATOR.html [a88707c84b4e]

↓

[Phase 2a] CI/CD

ci.yml [1051bec5d7df]

deploy.yml [69f608683b92]

build.yml [existing]

[Phase 2b] Scripts

compute_hashes.py [8760490835ee]

validate_links.py [2f25e5859101]

release.sh [3ab21c223371]

↓

[Phase 3] Artifact Generation → Data from config.yaml

hero pages (3)

deploy configs (2)

↓

[Phase 4] Validation → Feedback Loop (22/22 tests pass)

↓

[Phase 5] Packaging → BUILD_LOG.md + QUICK_ACCESS.md

↓

[Phase 6] Git Commit → Tagged release

Recursive Hooks

Phase	Triggers	Triggered By	Feedback Loop
Phase 0	Phase 1	User request	Initial state audit
Phase 1	Phase 2a, 2b	Phase 0	Landing page regen, archive validation
Phase 2a	Phase 3	Phase 1	CI/CD pipeline creation
Phase 2b	Phase 3	Phase 1	Script creation
Phase 3	Phase 4	Phase 2a, 2b	Hero pages, deploy configs
Phase 4	Phase 5 OR Phase 1	Phase 3	Test failures → fix → retest
Phase 5	Phase 6	Phase 4	Build log, quick access
Phase 6	—	Phase 5	Git commit

[2026-05-14T11:30:00Z] Phase 0: Setup & State Audit

Triggered by: User request for comprehensive implementation

Input artifacts:

- data/config.yaml [SHA-256: f49b9565cee0...] — SSoT v4.0.0
- docs/manifest.json [SHA-256: (pre-update)] — Build manifest
- VERSION → 4.0.0

Actions:

1. Audited project structure — 18 directories, 205+ files
2. Verified git state — commit fc31710 on main
3. Confirmed v4.0.0 with 3 compressors, SSoT config.yaml
4. Identified 10 implementation items from recommendations
5. Checked existing CI/CD — build.yml present, ci.yml / deploy.yml missing

Observations:

- index_v3_1.html archived → tests referencing it need update
- docs/manifest.json files had empty hashes → needs SHA-256 population
- No scripts/ directory existed → creating from scratch
- Hero pages not yet generated
- Link checker not present

Recursive hooks:

- Triggers: Phase 1 (all immediate fixes)
- ← Triggered by: User request

Validation:  Project state understood, dependency tree mapped

[2026-05-14T11:35:00Z] Phase 1: Immediate Fixes

Triggered by: Phase 0 audit findings

Input artifacts:

- `src/build_dashboard.py` [SHA-256: a3ced8007195...]
- `docs/NAVIGATOR.html` [SHA-256: (pre-update)]
- `docs/manifest.json` [SHA-256: (pre-update)]
- `tests/test_outputs.py` [SHA-256: (pre-update)]

Task 1.1: Regenerate Landing Page

Actions:

1. Ran `python3 src/build_dashboard.py`
2. Verified output shows v4.0.0
3. Confirmed links point to `index_v4_0.html` (not `index_v3_1.html`)

Output: `docs/index.html` [SHA-256: 93b94be27890...]

Task 1.2: Archive Stale Files

Actions:

1. Verified `docs/archive/` contains:
 - `index_v3_0_0_ARCHIVE.html`
 - `index_v3_1_0_ARCHIVE.html`
 - `presentation_v2_5_0_ARCHIVE.html`
 - `README.md` (deprecation notice)
2. Confirmed `STALE_FILES_AUDIT.md` documents all archived items

Output: Archive structure verified 

Task 1.3: Update NAVIGATOR.html

Actions:

1. Added "Audience Hero Pages" section with 3 cards
2. Added "CI/CD & Automation" section with 6 cards
3. All cards have status pills and descriptions

Output: `docs/NAVIGATOR.html` [SHA-256: a88707c84b4e...]

Task 1.4: Update Manifest with SHA-256

Actions:

1. Populated all 205 file entries with computed SHA-256 hashes
2. Updated file sizes

Output: `docs/manifest.json` [SHA-256: 7f6963680ad5...]

Task 1.5: Fix Test References

Actions:

1. Updated `tests/test_outputs.py` : `index_v3_1.html` → `index_v4_0.html`
2. Added `NAVIGATOR.html` to expected files list
3. Verified all 22 tests pass

Output: `tests/test_outputs.py` [SHA-256: updated]

Recursive hooks:

- Triggers: Phase 2a (CI/CD), Phase 2b (Scripts)
- ← Triggered by: Phase 0

Validation:  Landing page regenerated, NAVIGATOR updated, 22/22 tests pass

[2026-05-14T11:45:00Z] Phase 2a: CI/CD Pipeline

Triggered by: Phase 1 completion

Input artifacts:

- `.github/workflows/build.yml` [existing]

Actions:

1. Created `.github/workflows/ci.yml` — continuous integration
 - Python 3.11 setup
 - pytest with verbose output
 - `validate.sh --quick`
 - Link checking via `validate_links.py`
 - Artifact upload (test report + broken links)
2. Created `.github/workflows/deploy.yml` — GitHub Pages deployment
 - Triggered on push to main
 - Full build → upload Pages artifact → deploy
 - Concurrency protection
3. Preserved existing `build.yml` (unchanged)

Output artifacts:

- `.github/workflows/ci.yml` [SHA-256: 1051bec5d7df...]
- `.github/workflows/deploy.yml` [SHA-256: 69f608683b92...]

Recursive hooks:

- Triggers: Phase 3 (artifact generation)
- ← Triggered by: Phase 1

Validation:  YAML syntax valid, workflow structure complete

[2026-05-14T11:50:00Z] Phase 2b: Automation Scripts

Triggered by: Phase 1 completion

Actions:

scripts/compute_hashes.py

- Recursive SHA-256 scanner for all project files
- Skips `.git`, `__pycache__`, `venv`, `htmlcov`
- JSON output with timestamp, file count, per-file metadata
- `--manifest` flag to write `docs/manifest_hashes.json`

Output: `scripts/compute_hashes.py` [SHA-256: 8760490835ee...]

scripts/validate_links.py

- Scans all HTML files for `href` and `src` attributes
- Classifies links (internal, external, anchor, data, mailto)
- Validates internal links exist on filesystem
- `--report` flag for JSON report to `dist/broken_links_report.json`
- Exit code 1 if broken links found (CI-friendly)

Output: `scripts/validate_links.py` [SHA-256: 2f25e5859101...]

scripts/release.sh

- VERSION file update
- Source hash computation
- Full build execution
- Output hash computation
- Release notes generation
- Git commit + tag

Output: `scripts/release.sh` [SHA-256: 3ab21c223371...]

scripts/deploy.sh

- One-command: build → validate → link check → package → hash
- Summary with version, commit, branch info

Output: `scripts/deploy.sh` [SHA-256: 8d7446a3d713...]

scripts/export_pdfs.sh

- PDF export via weasyprint (graceful fallback)
- Exports: landing page, executive summary, dashboard, RTM

Output: `scripts/export_pdfs.sh` [SHA-256: computed]

Recursive hooks:

→ Triggers: Phase 3 (used by artifact generation)

← Triggered by: Phase 1

Validation:  All scripts executable, hash computation verified

[2026-05-14T12:00:00Z] Phase 3: Automated Artifact Generation

Triggered by: Phase 2a, 2b completion

Input artifacts:

- data/config.yaml [SHA-256: f49b9565cee0...]
- data/leak_classes.json
- data/valve_candidates.json
- data/scenarios.json
- data/source_anchors.json

Task 3.1: Hero Pages

Actions:

1. Created src/generate_hero_pages.py — reads SSoT config.yaml
2. Generated 3 audience-specific landing pages:
 - docs/heroes/executive.html — ROI / cost / valve derogation
 - docs/heroes/technical.html — leak classes / pressure / calculation engine
 - docs/heroes/compliance.html — EN 13185 / RTM-048 / PED
3. All pages responsive, styled, with navigation links

Output: 3 hero HTML files in docs/heroes/

Task 3.2: Deployment Configurations

Actions:

1. Created netlify.toml — build command, redirect to NAVIGATOR, security headers
2. Created vercel.json — build command, rewrite rules, security headers

Output artifacts:

- netlify.toml [SHA-256: 5d1dc282a940...]
- vercel.json [SHA-256: 97812dc107ee...]

Recursive hooks:

- Triggers: Phase 4 (validation)
- ← Triggered by: Phase 2a, 2b

Validation:  Hero pages generated, deployment configs valid

[2026-05-14T12:05:00Z] Phase 4: Comprehensive Validation

Triggered by: Phase 3 completion

Actions:

1. **Test suite:** 22/22 tests passing
 - test_config_loader.py — 4/4 
 - test_data_integrity.py — 3/3 
 - test_engineering.py — 5/5 
 - test_outputs.py — 3/3 
 - test_build_outputs.py — 2/2 
 - test_scenarios.py — 5/5 
1. **Link validation:** 84 HTML files scanned
 - 407 total links found
 - 310 internal links validated
 - 70 external links (skipped)

- 27 anchor links (OK)
- 7 broken internal links (legacy v3.x references in archived pages)

2. **Hash computation:** All source files SHA-256 computed

Feedback loop:

- Test failures in `test_outputs.py` → fixed v3.1 references → retest → pass
- Manifest missing hashes → populated → `test_build_outputs.py` → pass

Recursive hooks:

- Triggers: Phase 5 (packaging)
- ← Triggered by: Phase 3
- ← Feedback to: Phase 1 (test fix triggered re-run)

Validation:  All tests pass (22/22), link report generated

[2026-05-14T12:10:00Z] Phase 5: Documentation & Packaging

Triggered by: Phase 4 validation

Actions:

1. Created `BUILD_LOG.md` (this file) — recursive build sequence
2. Created `QUICK_ACCESS.md` — all URLs and quick commands
3. All artifacts tracked with SHA-256 hashes

Output artifacts:

- `BUILD_LOG.md` — recursive build log
- `QUICK_ACCESS.md` — quick access URLs and commands

Recursive hooks:

- Triggers: Phase 6 (git commit)
- ← Triggered by: Phase 4

[2026-05-14T12:15:00Z] Phase 6: Git Commit

Triggered by: Phase 5 completion

Actions:

1. `git add -A` — stage all new/modified files
2. `git commit` — comprehensive commit message
3. All artifacts versioned

Timeline

Forward Flow (Chronological)

```
[2026-05-14T11:30:00Z] → Phase 0: Audit & Setup
[2026-05-14T11:35:00Z] → Phase 1: Immediate Fixes (landing, navigator, tests)
[2026-05-14T11:45:00Z] → Phase 2a: CI/CD Pipelines
[2026-05-14T11:50:00Z] → Phase 2b: Automation Scripts
[2026-05-14T12:00:00Z] → Phase 3: Hero Pages & Deploy Configs
[2026-05-14T12:05:00Z] → Phase 4: Validation (22/22 tests pass)
[2026-05-14T12:10:00Z] → Phase 5: BUILD_LOG + QUICK_ACCESS
[2026-05-14T12:15:00Z] → Phase 6: Git Commit
```

Backward Flow (Causality)

```
Phase 6 (git)
  ← Phase 5 (BUILD_LOG, QUICK_ACCESS)
    ← Phase 4 (22/22 tests, link validation)
      ← Phase 3 (hero pages, deploy configs)
        ← Phase 2a (CI/CD) + Phase 2b (scripts)
          ← Phase 1 (landing, navigator, manifest, test fixes)
            ← Phase 0 (audit)
              ← User request
```

Artifact Manifest (SHA-256 Provenance)

Artifact	SHA-256 (first 12)	Description
data/config.yaml	f49b9565cee0	SSoT configuration
src/calc_leak_rate.py	347c7215e084	Physics engine (no alignment factors)
src/generate_dashboard.py	d3351f31c363	Dashboard HTML generator
src/build_dashboard.py	a3ced8007195	Landing page builder
src/generate_hero_pages.py	(new)	Audience hero page generator
docs/index.html	93b94be27890	Landing page (v4.0.0)
docs/NAVIGATOR.html	a88707c84b4e	Comprehensive navigator
docs/index_v4_0.html	826e1031319c	40-slide master navigator
docs/dashboard.html	1b23f6441181	Interactive Plotly dashboard
docs/manifest.json	7f6963680ad5	Build manifest (205 files)
scripts/compute_hashes.py	8760490835ee	SHA-256 file scanner
scripts/validate_links.py	2f25e5859101	HTML link validator
scripts/release.sh	3ab21c223371	Release automation
scripts/deploy.sh	8d7446a3d713	One-command deploy
.github/workflows/ci.yml	1051bec5d7df	CI pipeline
.github/workflows/deploy.yml	69f608683b92	Pages deployment
netlify.toml	5d1dc282a940	Netlify configuration
vercel.json	97812dc107ee	Vercel configuration

Deliverables Checklist

- [x] BUILD_LOG.md (this file — recursive sequence with SHA-256)
- [x] .github/workflows/ci.yml (CI pipeline)
- [x] .github/workflows/deploy.yml (Pages deployment)

- [x] `scripts/release.sh` (versioning with hashes)
- [x] `scripts/compute_hashes.py` (SHA-256 scanner)
- [x] `scripts/validate_links.py` (link checker)
- [x] `src/generate_hero_pages.py` (audience hero pages)
- [x] `scripts/export_pdfs.sh` (PDF generation)
- [x] `netlify.toml`, `vercel.json` (deployment configs)
- [x] `QUICK_ACCESS.md` (URLs and commands)
- [x] `scripts/deploy.sh` (one-command deploy)
- [x] Updated `docs/NAVIGATOR.html` (hero + CI/CD sections)
- [x] Updated `docs/manifest.json` (SHA-256 populated)
- [x] Updated `tests/test_outputs.py` (v4.0.0 references)
- [x] All 22 tests passing
- [x] Git commit with timestamp